

```

=====
57.3/57.3 kB 2.5 MB/s eta 0:00:00
5.4/5.4 MB 59.3 MB/s eta 0:00:00
937.5/937.5 kB 58.9 MB/s eta 0:00:00
25.5/25.5 MB 20.2 MB/s eta 0:00:00
51.5/51.5 kB 3.0 MB/s eta 0:00:00
2.4/2.4 MB 40.2 MB/s eta 0:00:00
8.8/8.8 MB 63.3 MB/s eta 0:00:00
167.2/167.2 kB 11.1 MB/s eta 0:00:00
=====
Using Device: cuda
=====
Mounted at /content/drive

Dataset found at:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification

Seeds:
[42, 52, 62, 72, 82]

Dataset paths:
Train: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/train
Validation: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/val
Test: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/test

Class mapping:
{'fractured': 0, 'not fractured': 1}

Dataset sizes:
Train      : 9246
Validation : 829
Test       : 506

Data loaders created successfully.

Quantum configuration:
Qubits      : 4
Quantum layers : 2
Weight shape : (2, 4, 3)
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0
100%|██████████| 20.5M/20.5M [00:00<00:00, 124MB/s]

=====
MODEL PARAMETER INFORMATION
=====
Total parameters      : 4,012,701
Trainable parameters  : 4,012,701
Compression parameters: 5,124
Quantum parameters    : 24
Output parameters     : 5

=====
STARTING FIVE-SEED EXPERIMENT — SEED 42
=====

Random Seed: 42

=====
STARTING DVQC TRAINING FOR SEED 42
=====
Seed 42 | Epoch 1/10: 100%|██████████| 145/145 [13:25<00:00, 5.56s/it, loss=0.598]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 1/10 | Train Loss: 0.6283 | Val Loss: 0.5972 | Train Acc: 84.88% | Val Acc: 96.98%
Seed 42 | Epoch 2/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.575]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 2/10 | Train Loss: 0.5957 | Val Loss: 0.5811 | Train Acc: 99.28% | Val Acc: 98.31%
Seed 42 | Epoch 3/10: 100%|██████████| 145/145 [05:04<00:00, 2.10s/it, loss=0.602]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 3/10 | Train Loss: 0.5813 | Val Loss: 0.5679 | Train Acc: 99.64% | Val Acc: 98.55%
Seed 42 | Epoch 4/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.574]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 4/10 | Train Loss: 0.5674 | Val Loss: 0.5551 | Train Acc: 99.75% | Val Acc: 98.67%
Seed 42 | Epoch 5/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.538]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 5/10 | Train Loss: 0.5531 | Val Loss: 0.5409 | Train Acc: 99.84% | Val Acc: 99.16%
Seed 42 | Epoch 6/10: 100%|██████████| 145/145 [04:56<00:00, 2.04s/it, loss=0.534]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 6/10 | Train Loss: 0.5392 | Val Loss: 0.5276 | Train Acc: 99.84% | Val Acc: 99.16%
Seed 42 | Epoch 7/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.518]

```

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 7/10 | Train Loss: 0.5251 | Val Loss: 0.5154 | Train Acc: 99.90% | Val Acc: 98.79%
Seed 42 | Epoch 8/10: 100%|██████████| 145/145 [04:56<00:00, 2.05s/it, loss=0.508]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 8/10 | Train Loss: 0.5113 | Val Loss: 0.5006 | Train Acc: 99.81% | Val Acc: 99.40%
Seed 42 | Epoch 9/10: 100%|██████████| 145/145 [04:53<00:00, 2.03s/it, loss=0.485]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 9/10 | Train Loss: 0.4978 | Val Loss: 0.4872 | Train Acc: 99.71% | Val Acc: 99.40%
Seed 42 | Epoch 10/10: 100%|██████████| 145/145 [05:03<00:00, 2.09s/it, loss=0.475]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 10/10 | Train Loss: 0.4834 | Val Loss: 0.4741 | Train Acc: 99.77% | Val Acc: 99.16%
```

Seed 42 completed in 62.27 minutes.

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

RESULTS FOR SEED 42

```
-----
Test Accuracy   : 99.01%
Test Sensitivity: 99.63%
Test Specificity: 98.32%
Test Precision  : 98.52%
Test F1-Score   : 99.07%
Test Error Rate : 0.99%
Test ROC-AUC    : 0.9999
```

STARTING FIVE-SEED EXPERIMENT – SEED 52

Random Seed: 52

STARTING DVQC TRAINING FOR SEED 52

```
=====
Seed 52 | Epoch 1/10: 100%|██████████| 145/145 [05:01<00:00, 2.08s/it, loss=0.574]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 1/10 | Train Loss: 0.6043 | Val Loss: 0.5694 | Train Acc: 92.60% | Val Acc: 95.42%
Seed 52 | Epoch 2/10: 100%|██████████| 145/145 [04:59<00:00, 2.07s/it, loss=0.564]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 2/10 | Train Loss: 0.5610 | Val Loss: 0.5510 | Train Acc: 99.21% | Val Acc: 97.71%
Seed 52 | Epoch 3/10: 100%|██████████| 145/145 [05:05<00:00, 2.10s/it, loss=0.559]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 3/10 | Train Loss: 0.5470 | Val Loss: 0.5359 | Train Acc: 99.60% | Val Acc: 98.55%
Seed 52 | Epoch 4/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.537]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 4/10 | Train Loss: 0.5354 | Val Loss: 0.5246 | Train Acc: 99.62% | Val Acc: 99.16%
Seed 52 | Epoch 5/10: 100%|██████████| 145/145 [04:57<00:00, 2.04s/it, loss=0.51]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 5/10 | Train Loss: 0.5238 | Val Loss: 0.5125 | Train Acc: 99.69% | Val Acc: 99.40%
Seed 52 | Epoch 6/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.504]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 6/10 | Train Loss: 0.5128 | Val Loss: 0.5021 | Train Acc: 99.76% | Val Acc: 99.40%
Seed 52 | Epoch 7/10: 100%|██████████| 145/145 [05:02<00:00, 2.09s/it, loss=0.508]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 7/10 | Train Loss: 0.5020 | Val Loss: 0.4919 | Train Acc: 99.83% | Val Acc: 99.40%
Seed 52 | Epoch 8/10: 100%|██████████| 145/145 [05:01<00:00, 2.08s/it, loss=0.491]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 8/10 | Train Loss: 0.4917 | Val Loss: 0.4815 | Train Acc: 99.84% | Val Acc: 99.52%
Seed 52 | Epoch 9/10: 100%|██████████| 145/145 [05:04<00:00, 2.10s/it, loss=0.471]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 9/10 | Train Loss: 0.4820 | Val Loss: 0.4731 | Train Acc: 99.83% | Val Acc: 99.40%
Seed 52 | Epoch 10/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.469]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 10/10 | Train Loss: 0.4722 | Val Loss: 0.4676 | Train Acc: 99.86% | Val Acc: 98.55%
```

Seed 52 completed in 52.17 minutes.

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
```

```
warnings.warn(
```

```
RESULTS FOR SEED 52
```

```
-----
Test Accuracy   : 97.23%
Test Sensitivity: 96.27%
Test Specificity: 98.32%
Test Precision  : 98.47%
Test F1-Score   : 97.36%
Test Error Rate : 2.77%
Test ROC-AUC    : 0.9921
```

```
=====
STARTING FIVE-SEED EXPERIMENT - SEED 62
=====
```

Random Seed: 62

```
=====
STARTING DVQC TRAINING FOR SEED 62
=====
```

```
Seed 62 | Epoch 1/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.639]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 1/10 | Train Loss: 0.6381 | Val Loss: 0.5652 | Train Acc: 50.18% | Val Acc: 59.35%
Seed 62 | Epoch 2/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.591]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 2/10 | Train Loss: 0.5795 | Val Loss: 0.5347 | Train Acc: 50.18% | Val Acc: 59.35%
Seed 62 | Epoch 3/10: 100%|██████████| 145/145 [04:58<00:00, 2.06s/it, loss=0.559]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 3/10 | Train Loss: 0.5575 | Val Loss: 0.5189 | Train Acc: 50.18% | Val Acc: 59.35%
Seed 62 | Epoch 4/10: 100%|██████████| 145/145 [04:59<00:00, 2.06s/it, loss=0.548]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 4/10 | Train Loss: 0.5409 | Val Loss: 0.5029 | Train Acc: 50.18% | Val Acc: 59.35%
Seed 62 | Epoch 5/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.495]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 5/10 | Train Loss: 0.5256 | Val Loss: 0.4896 | Train Acc: 83.13% | Val Acc: 98.79%
Seed 62 | Epoch 6/10: 100%|██████████| 145/145 [04:59<00:00, 2.07s/it, loss=0.553]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 6/10 | Train Loss: 0.5102 | Val Loss: 0.4783 | Train Acc: 99.13% | Val Acc: 98.91%
Seed 62 | Epoch 7/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.459]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 7/10 | Train Loss: 0.4948 | Val Loss: 0.4664 | Train Acc: 99.38% | Val Acc: 98.91%
Seed 62 | Epoch 8/10: 100%|██████████| 145/145 [04:58<00:00, 2.06s/it, loss=0.447]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 8/10 | Train Loss: 0.4793 | Val Loss: 0.4517 | Train Acc: 99.70% | Val Acc: 99.03%
Seed 62 | Epoch 9/10: 100%|██████████| 145/145 [05:01<00:00, 2.08s/it, loss=0.436]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 9/10 | Train Loss: 0.4647 | Val Loss: 0.4380 | Train Acc: 99.69% | Val Acc: 99.28%
Seed 62 | Epoch 10/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.463]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 62 | Epoch 10/10 | Train Loss: 0.4514 | Val Loss: 0.4255 | Train Acc: 99.43% | Val Acc: 99.40%
```

Seed 62 completed in 51.89 minutes.

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

```
RESULTS FOR SEED 62
```

```
-----
Test Accuracy   : 99.41%
Test Sensitivity: 98.88%
Test Specificity: 100.00%
Test Precision  : 100.00%
Test F1-Score   : 99.44%
Test Error Rate : 0.59%
Test ROC-AUC    : 0.9942
```

```
=====
STARTING FIVE-SEED EXPERIMENT - SEED 72
=====
```

Random Seed: 72

```
=====
STARTING DVQC TRAINING FOR SEED 72
=====
```

```
Seed 72 | Epoch 1/10: 100%|██████████| 145/145 [05:02<00:00, 2.08s/it, loss=0.64]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 1/10 | Train Loss: 0.6118 | Val Loss: 0.5886 | Train Acc: 91.50% | Val Acc: 96.86%
Seed 72 | Epoch 2/10: 100%|██████████| 145/145 [05:02<00:00, 2.09s/it, loss=0.552]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 2/10 | Train Loss: 0.5640 | Val Loss: 0.5665 | Train Acc: 99.46% | Val Acc: 98.55%
Seed 72 | Epoch 3/10: 100%|██████████| 145/145 [05:02<00:00, 2.08s/it, loss=0.537]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 3/10 | Train Loss: 0.5482 | Val Loss: 0.5507 | Train Acc: 99.70% | Val Acc: 99.03%
Seed 72 | Epoch 4/10: 100%|██████████| 145/145 [05:03<00:00, 2.09s/it, loss=0.532]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 4/10 | Train Loss: 0.5352 | Val Loss: 0.5365 | Train Acc: 99.77% | Val Acc: 99.16%
Seed 72 | Epoch 5/10: 100%|██████████| 145/145 [05:07<00:00, 2.12s/it, loss=0.512]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 5/10 | Train Loss: 0.5227 | Val Loss: 0.5238 | Train Acc: 99.78% | Val Acc: 99.16%
Seed 72 | Epoch 6/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.505]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 6/10 | Train Loss: 0.5098 | Val Loss: 0.5089 | Train Acc: 99.87% | Val Acc: 99.64%
Seed 72 | Epoch 7/10: 100%|██████████| 145/145 [05:01<00:00, 2.08s/it, loss=0.49]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 7/10 | Train Loss: 0.4971 | Val Loss: 0.4959 | Train Acc: 99.90% | Val Acc: 99.52%
Seed 72 | Epoch 8/10: 100%|██████████| 145/145 [05:01<00:00, 2.08s/it, loss=0.485]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 8/10 | Train Loss: 0.4849 | Val Loss: 0.4835 | Train Acc: 99.86% | Val Acc: 99.28%
Seed 72 | Epoch 9/10: 100%|██████████| 145/145 [05:04<00:00, 2.10s/it, loss=0.467]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 9/10 | Train Loss: 0.4723 | Val Loss: 0.4697 | Train Acc: 99.90% | Val Acc: 99.64%
Seed 72 | Epoch 10/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.451]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 10/10 | Train Loss: 0.4602 | Val Loss: 0.4554 | Train Acc: 99.85% | Val Acc: 99.88%
```

```
Seed 72 completed in 52.53 minutes.
```

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

```
RESULTS FOR SEED 72
-----
```

```
Test Accuracy : 99.01%
Test Sensitivity: 99.63%
Test Specificity: 98.32%
Test Precision : 98.52%
Test F1-Score : 99.07%
Test Error Rate : 0.99%
Test ROC-AUC : 0.9980
```

```
=====
STARTING FIVE-SEED EXPERIMENT - SEED 82
=====
```

```
Random Seed: 82
```

```
=====
STARTING DVQC TRAINING FOR SEED 82
=====
```

```
Seed 82 | Epoch 1/10: 100%|██████████| 145/145 [05:00<00:00, 2.08s/it, loss=0.58]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 1/10 | Train Loss: 0.6191 | Val Loss: 0.5951 | Train Acc: 80.61% | Val Acc: 92.52%
Seed 82 | Epoch 2/10: 100%|██████████| 145/145 [05:00<00:00, 2.07s/it, loss=0.543]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 2/10 | Train Loss: 0.5502 | Val Loss: 0.5580 | Train Acc: 98.26% | Val Acc: 97.35%
Seed 82 | Epoch 3/10: 100%|██████████| 145/145 [04:58<00:00, 2.06s/it, loss=0.514]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 3/10 | Train Loss: 0.5269 | Val Loss: 0.5396 | Train Acc: 99.24% | Val Acc: 97.95%
Seed 82 | Epoch 4/10: 100%|██████████| 145/145 [04:57<00:00, 2.05s/it, loss=0.498]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

```
07/09/2026, 18:41                                EFFICIENTNET-B0 + DVQC n=4.ipynb - Colab

Seed 82 | Epoch 4/10 | Train Loss: 0.5116 | Val Loss: 0.5237 | Train Acc: 99.28% | Val Acc: 98.19%
Seed 82 | Epoch 5/10: 100%[REDACTED] | 145/145 [05:00<00:00, 2.08s/it, loss=0.49]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 5/10 | Train Loss: 0.4965 | Val Loss: 0.5078 | Train Acc: 99.72% | Val Acc: 99.03%
Seed 82 | Epoch 6/10: 100%[REDACTED] | 145/145 [05:01<00:00, 2.08s/it, loss=0.468]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 6/10 | Train Loss: 0.4836 | Val Loss: 0.4931 | Train Acc: 99.63% | Val Acc: 99.28%
Seed 82 | Epoch 7/10: 100%[REDACTED] | 145/145 [04:59<00:00, 2.07s/it, loss=0.494]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 7/10 | Train Loss: 0.4706 | Val Loss: 0.4820 | Train Acc: 99.70% | Val Acc: 99.03%
Seed 82 | Epoch 8/10: 100%[REDACTED] | 145/145 [05:02<00:00, 2.08s/it, loss=0.456]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 8/10 | Train Loss: 0.4583 | Val Loss: 0.4655 | Train Acc: 99.62% | Val Acc: 99.40%
Seed 82 | Epoch 9/10: 100%[REDACTED] | 145/145 [05:01<00:00, 2.08s/it, loss=0.434]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 9/10 | Train Loss: 0.4453 | Val Loss: 0.4565 | Train Acc: 99.66% | Val Acc: 98.19%
Seed 82 | Epoch 10/10: 100%[REDACTED] | 145/145 [05:04<00:00, 2.10s/it, loss=0.431]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 10/10 | Train Loss: 0.4319 | Val Loss: 0.4380 | Train Acc: 99.63% | Val Acc: 99.28%
```

```
Seed 82 completed in 52.24 minutes.
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

RESULTS FOR SEED 82

Test Accuracy : 98.62%
Test Sensitivity: 98.88%
Test Specificity: 98.32%
Test Precision : 98.51%
Test F1-Score : 98.70%
Test Error Rate : 1.38%
Test ROC-AUC : 0.9905

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INDIVIDUAL FIVE-SEED RESULTS

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	Seed	Split	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	Error Rate (%)	ROC-AUC
0	42	Training	99.945923	99.978448	99.913157	99.913849	99.946138	0.054077	0.999997
1	42	Validation	99.155609	99.796748	98.219585	98.792757	99.292214	0.844391	0.999717
2	42	Testing	99.011858	99.626866	98.319328	98.523985	99.072356	0.988142	0.999875
3	52	Training	99.891845	99.784483	100.000000	100.000000	99.892125	0.108155	0.999625
4	52	Validation	98.552473	97.967480	99.406528	99.586777	98.770492	1.447527	0.996683
5	52	Testing	97.233202	96.268657	98.319328	98.473282	97.358491	2.766798	0.992083
6	62	Training	99.783690	99.935345	99.630916	99.634723	99.784807	0.216310	0.999502
7	62	Validation	99.396864	99.390244	99.406528	99.592668	99.491353	0.603136	0.996568
8	62	Testing	99.407115	98.880597	100.000000	100.000000	99.437148	0.592885	0.994207
9	72	Training	99.989185	99.978448	100.000000	100.000000	99.989223	0.010815	0.999893
10	72	Validation	99.879373	99.796748	100.000000	100.000000	99.898271	0.120627	0.998758
11	72	Testing	99.011858	99.626866	98.319328	98.523985	99.072356	0.988142	0.998025
12	82	Training	99.881030	99.806034	99.956578	99.956831	99.881376	0.118970	0.999839
13	82	Validation	99.276236	99.390244	99.109792	99.390244	99.390244	0.723764	0.998727
14	82	Testing	98.616601	98.880597	98.319328	98.513011	98.696462	1.383399	0.990531

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DVQC FIVE-SEED MEAN ± STANDARD DEVIATION

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	Split	Metric	Mean	Standard Deviation	Mean ± SD
0	Training	Accuracy (%)	99.898334	0.077465	99.90 ± 0.08
1	Training	Sensitivity (%)	99.896552	0.094435	99.90 ± 0.09

	Split	Metric	Mean	Standard Deviation	Mean ± SD
2	Training	Specificity (%)	99.900130	0.154742	99.90 ± 0.15
3	Training	Precision (%)	99.901081	0.153124	99.90 ± 0.15
4	Training	F1-Score (%)	99.898734	0.077071	99.90 ± 0.08
5	Training	Error Rate (%)	0.101666	0.077465	0.10 ± 0.08
6	Training	ROC-AUC	0.999771	0.000203	1.00 ± 0.00
7	Validation	Accuracy (%)	99.252111	0.477964	99.25 ± 0.48
8	Validation	Sensitivity (%)	99.268293	0.755048	99.27 ± 0.76
9	Validation	Specificity (%)	99.228487	0.650116	99.23 ± 0.65
10	Validation	Precision (%)	99.472489	0.440049	99.47 ± 0.44
11	Validation	F1-Score (%)	99.368515	0.406096	99.37 ± 0.41
12	Validation	Error Rate (%)	0.747889	0.477964	0.75 ± 0.48
13	Validation	ROC-AUC	0.998091	0.001396	1.00 ± 0.00
14	Testing	Accuracy (%)	98.656126	0.843112	98.66 ± 0.84
15	Testing	Sensitivity (%)	98.656716	1.386133	98.66 ± 1.39
16	Testing	Specificity (%)	98.655462	0.751619	98.66 ± 0.75
17	Testing	Precision (%)	98.806853	0.667316	98.81 ± 0.67
18	Testing	F1-Score (%)	98.727363	0.808796	98.73 ± 0.81
19	Testing	Error Rate (%)	1.343874	0.843112	1.34 ± 0.84
20	Testing	ROC-AUC	0.994944	0.003937	0.99 ± 0.00

=====
FINAL TEST-SET MEAN ± STANDARD DEVIATION
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	Metric	Mean	Standard Deviation	Mean ± SD
14	Accuracy (%)	98.656126	0.843112	98.66 ± 0.84
15	Sensitivity (%)	98.656716	1.386133	98.66 ± 1.39
16	Specificity (%)	98.655462	0.751619	98.66 ± 0.75
17	Precision (%)	98.806853	0.667316	98.81 ± 0.67
18	F1-Score (%)	98.727363	0.808796	98.73 ± 0.81
19	Error Rate (%)	1.343874	0.843112	1.34 ± 0.84
20	ROC-AUC	0.994944	0.003937	0.99 ± 0.00

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TEST CONFUSION MATRICES FOR FIVE SEEDS
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Seed 42
[[234 4]
[1 267]]

Seed 52
[[234 4]
[10 258]]

Seed 62
[[238 0]
[3 265]]

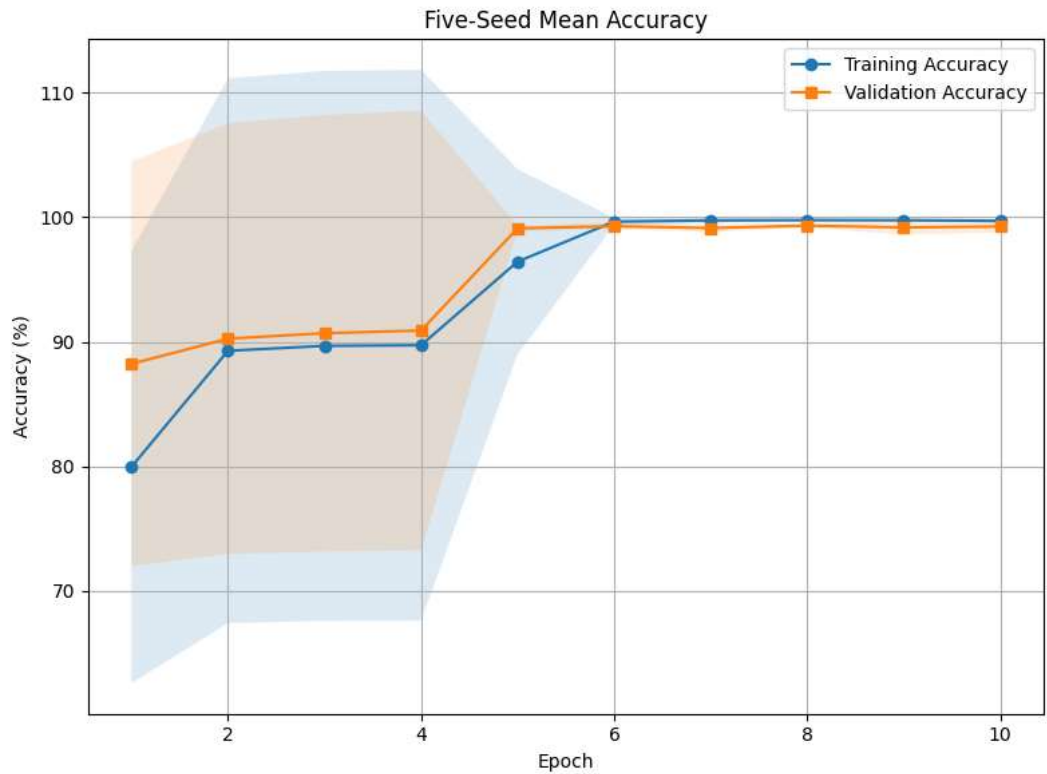
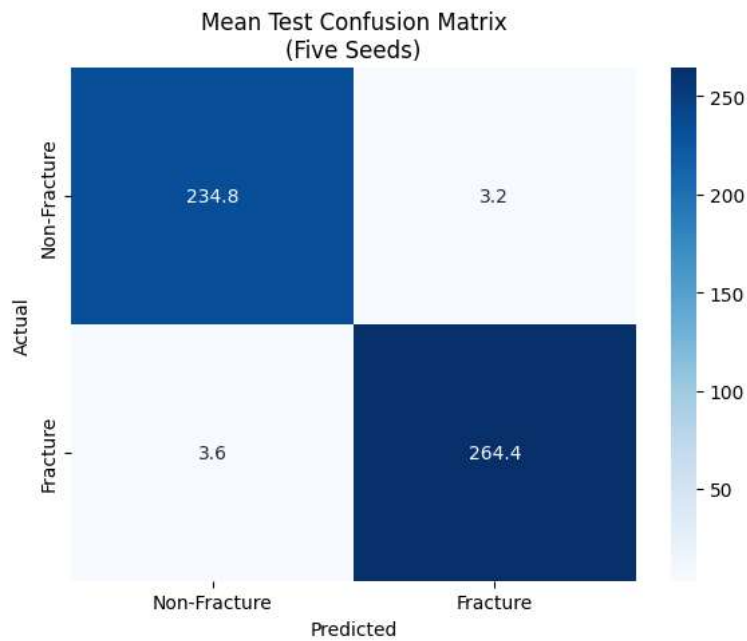
Seed 72
[[234 4]
[1 267]]

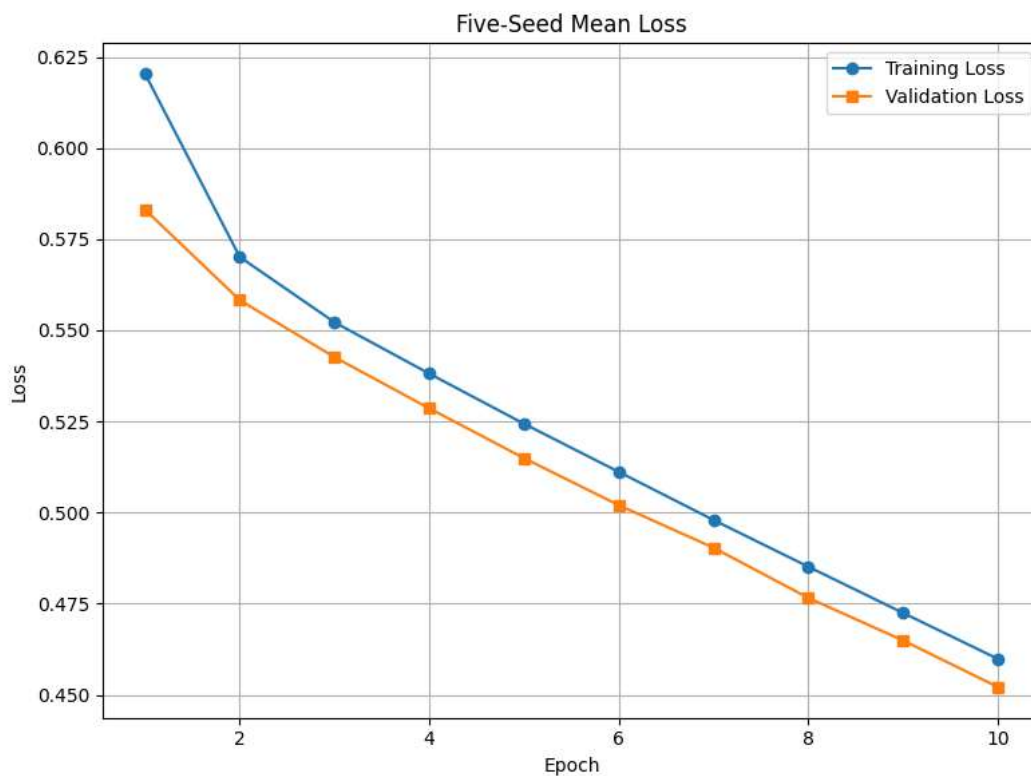
Seed 82
[[234 4]
[3 265]]

=====

MEAN TEST CONFUSION MATRIX

```
[[234.8  3.2]
 [ 3.6 264.4]]
```





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FINAL VERIFICATION

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✓ Five independent seeds used:
 [42, 52, 62, 72, 82]
 ✓ Training metrics calculated
 ✓ Validation metrics calculated
 ✓ Testing metrics calculated
 ✓ Accuracy calculated
 ✓ Sensitivity calculated
 ✓ Specificity calculated
 ✓ Precision calculated
 ✓ F1-Score calculated
 ✓ Error Rate calculated
 ✓ ROC-AUC calculated
 ✓ Mean calculated
 ✓ Standard Deviation calculated
 ✓ Mean $\pm$ SD calculated
 ✓ Individual seed results saved
 ✓ Mean $\pm$ SD results saved

Saved files:

1. DVQC_5Seed_Individual_Results.csv
2. DVQC_Mean_SD_Results.csv

Experiment completed successfully.